Terraform Infrastructure Guide

Infrastructure as Code

Terraform is an infrastructure-as-code tool that allows infrastructure to be defined in configuration files. The configuration describes the desired resources and their relationships.

Providers and Resources

Terraform providers allow Terraform to interact with platforms such as AWS. Resources represent infrastructure objects such as VPCs, subnets, IAM roles, and EKS clusters.

Terraform State

Terraform maintains state to track the infrastructure resources it manages. State allows Terraform to determine what changes are required when the configuration changes.

Plan and Apply

The terraform plan command previews proposed changes. The terraform apply command applies those changes. Reviewing a plan before applying it helps prevent unexpected infrastructure modifications.

Modules

Terraform modules group reusable infrastructure configurations. Modules can standardize common patterns such as VPC creation, EKS clusters, networking, and IAM configuration.